

B.Tech IV Year I Semester (R20) Regular & Supplementary Examinations December/January 2024/2025

RADAR ENGINEERING

(Electronics and Communication Engineering)

Time: 3 hours

Max. Marks: 70

PART – A

(Compulsory Question)

1 Answer the following: (10 X 02 = 20 Marks)

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| (a) List the radar frequencies. | 2M |
| (b) Explain Maximum Unambiguous Range. | 2M |
| (c) What is Doppler effect in radars? | 2M |
| (d) What is the disadvantage of CW radar? How to overcome it? | 2M |
| (e) What is the use of Staggered PRF? | 2M |
| (f) List the Limitations to MTI Performance. | 2M |
| (g) Define Noise temperature. | 2M |
| (h) Draw the Acquisition and Scanning Patterns of tracking radar. | 2M |
| (i) What is Direction Finder in navigation? | 2M |
| (j) Explain the need of Phased Array Antennas. | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

- 2 Explain the applications of radar. With neat diagram explain the concept of minimum detectable signal. 10M

OR

- 3 Discuss in detail the system losses in radar. Explain the concept of Integration of Radar Pulses. 10M

- 4 With neat block diagram explain the Range and Doppler Measurement in FMCW radar. Give the applications of FM-CW radar. 10M

OR

- 5 Compute the velocity of the target if it produces a Doppler shift of 1 KHz and operating wavelength is 3 cm. Derive the relationship used. 10M

- 6 With a neat schematic diagram, explain the operation of MTI radar using power amplifier. Explain blind speed in radars and give the remedy for the same. 10M

OR

- 7 Draw the block diagram and explain block by block about the Range Gated Doppler Filters for radar. 10M

- 8 Explain various limitations of tracking radars. Explain the monopulse amplitude tracking radar operation using a block diagram. 10M

OR

- 9 Explain different scanning techniques and tracking mechanisms. Explain about cross correlation receiver. 10M

- 10 Explain Navigational aids in detail. Write notes on Architecture for Phased Arrays. 10M

OR

- 11 Explain Frequency scan Arrays in radar. Discuss the basic concepts of Phased Array Antennas. 10M
